Understanding Contributions to Reductions in Cost of Transport with Stiffness-Customized Ankle-Foot Orthosis: A Case Series

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Background

Stiffness-customized passive-dynamic ankle-foot orthoses (PD-AFOs) enable individuals post-stroke to reduce their lower extremity mechanical cost of transport (COT) compared to walking with No AFO or with their standard-of-care (SOC) AFO [1], although how this COT reduction is achieved is unknown

Purpose:
Investigate how individuals achieve this ↓ COT

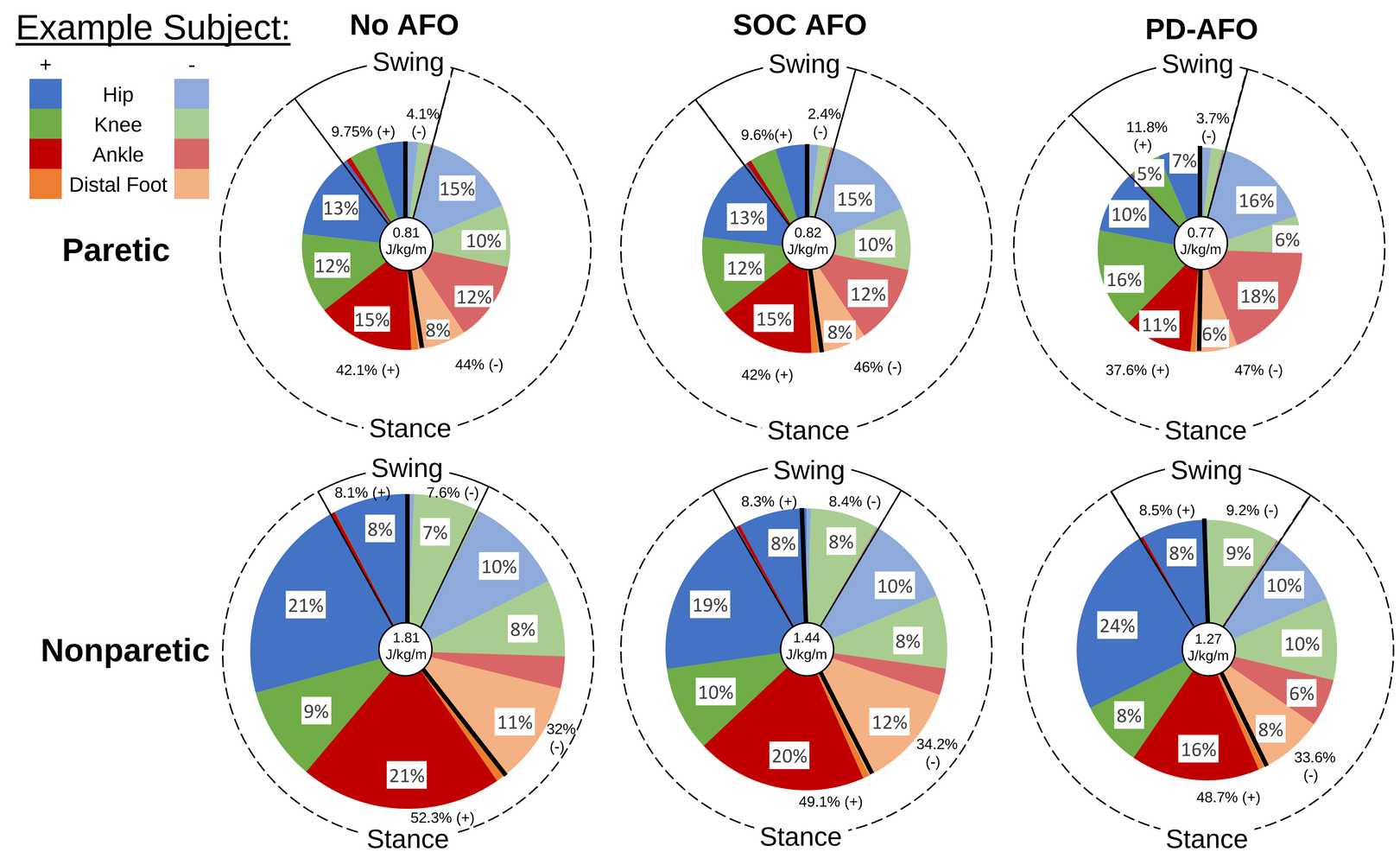
Hypothesis:
↓ COT is driven primarily by ↓ paretic ankle work

Methods

- Eight individuals six+ months post-stroke walked on an instrumented treadmill at their self-selected walking speed determined via 10m walk test with No AFO. Subjects also walked at the same speed wearing their SOC AFO
- Stiffness-customized PD-AFO was made using No AFO data and worn by subject for four weeks
- Subjects returned to lab to walk on the instrumented treadmill with the PD-AFO at the same No AFO speed
- CLEW approach was used to quantify and visualize how the energetics of each limb constituent changed between conditions [2]
- COT is the summation of each constituents' work:

$$COT = \frac{\frac{hip^{+} + knee^{+} + ankle^{+} + foot^{+}}{mass} + abs\left(\frac{hip^{-} + knee^{-} + ankle^{-} + foot^{-}}{mass}\right)}{stride\ length}$$

Results



- ↓ Total COT with the PD-AFO compared to No AFO **and** SOC AFO
- ↓ COT was **not** driven by the ankle, but rather due to a summation of small changes across numerous constituents of both limbs
- All subjects showed small changes in numerous constituents, although which constituents changed varied across subject

Conclusions

- PD-AFOs cause adjustments of much of the entire post-stroke lower limb system rather than simply direct enhancement of plantar flexor function

Acknowledgments

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References [1] Koller, C. et al. Proc. ASB Annual Meeting, 2021. [2] Ebrahimi, A. et al. Gait Posture, 56, 49-53, 2017.